Surface integrals

Any integral which is to be evaluated over a surface is called a surface integral.

The surface integral of a vector point function $\vec{F}$ express the flux through a surface.

If $\vec{F}$ represents the velocity of a fluid, then $\iint_S \vec{F}\cdot\hat{n}\,ds$ gives the flux across the surface S.

* If $\vec{F}$ represents gravitational force, electric force, magnetic force, in each case $\iint_S \vec{F}\cdot\hat{n}\,ds$ gives the corresponding flux.

To evaluate any surface integral, it is convenient to evaluate the double integral of its projection in xy or yz or xz plane.

Working Rule

(1) For any given surface ϕ, find $\hat{n} = \frac{\nabla\phi}{|\nabla\phi|}$

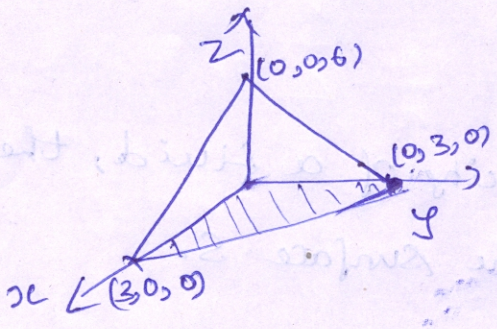
(2) Find $\vec{F}\cdot\hat{n}$

(3) If the projection of S is taken in xy plane, then $ds = \frac{dx\,dy}{|\hat{n}\cdot\hat{k}|}$, while $ds = \frac{dy\,dz}{|\hat{n}\cdot\hat{i}|}$ if the projection is in yz plane

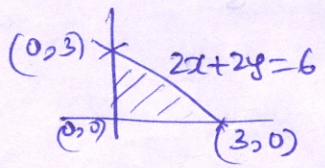
and $ds = \frac{dx\,dz}{|\hat{n}\cdot\hat{j}|}$ if the projection is in xz plane

① Evaluate $\iint_S \vec{F}\cdot\hat{n}\,ds$ where S is the surface of the plane $2x+2y+z=6$ included in the first octant and $\vec{F} = xy\hat{i} - x^2\hat{j} + (x+z)\hat{k}$.

Soln:



Project S on the xy plane. this projection of S on the xy plane is triangular region, bounded by x-axis, y-axis and $2x+2y=6$



let $\phi = 2x+2y+z-6$

$$\nabla\phi = \frac{\partial\phi}{\partial x}\hat{i} + \frac{\partial\phi}{\partial y}\hat{j} + \frac{\partial\phi}{\partial z}\hat{k}$$

$$= 2\hat{i} + 2\hat{j} + \hat{k}$$

$$|\nabla\phi| = \sqrt{2^2+2^2+1^2} = \sqrt{9} = 3$$

Unit normal vector $\hat{n}$ is given by

$$\hat{n} = \frac{\nabla\phi}{|\nabla\phi|} = \frac{2\hat{i}+2\hat{j}+\hat{k}}{3}$$

$$\vec{F}\cdot\hat{n} = (xy\hat{i} - x^2\hat{j} + (x+z)\hat{k})\cdot\frac{1}{3}(2\hat{i}+2\hat{j}+\hat{k})$$

$$= \frac{2xy - 2x^2 + x + z}{3}$$

$$\hat{n}\cdot\hat{k} = \frac{2\hat{i}+2\hat{j}+\hat{k}}{3}\cdot\hat{k} = \frac{1}{3}$$

Now, $\iint_S \vec{F}\cdot\hat{n}\,ds = \iint_R \vec{F}\cdot\hat{n}\,\frac{dx\,dy}{|\hat{n}\cdot\hat{k}|}$

$$= \int_{x=0}^{x=3}\int_{y=0}^{y=3-x} \frac{(2xy-2x^2+x+z)}{3}\,\frac{dy\,dx}{\frac{1}{3}}$$

$$= \int_0^3\int_0^{3-x} (2xy - 2x^2 + x + 6 - 2x - 2y)\,dy\,dx$$

$(\because 2x+2y+z=6)$

$\Rightarrow z = 6-2x-2y$

$$= \int_0^3\int_0^{3-x} (2xy - 2x^2 - x - 2y + 6)\,dy\,dx$$

$$= \int_0^3 \left[xy^2 - 2x^2y - xy - y^2 + 6y\right]_0^{3-x} dx$$

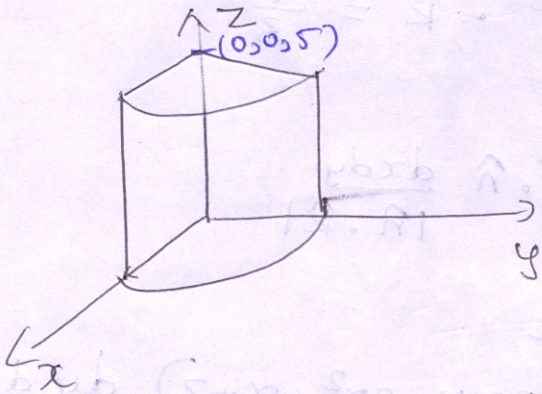
$$= \int_0^3 \left[x(3-x)^2 - 2x^2(3-x) - x(3-x) - (3-x)^2 + 6(3-x)\right] dx$$

$$= \int_0^3 (3x^3 - 12x^2 + 6x + 9)\,dx$$

$$= \frac{3}{4}x^4 - 4x^3 + 3x^2 + 9x\Big]_0^3 = 27/4$$

(2) Evaluate $\iint_S \vec{F} \cdot \hat{n}\, ds$ where $\vec{F} = z\hat{i} + x\hat{j} + 3y^2 z\hat{k}$ and S is the surface of the cylinder $x^2+y^2=16$ included in the first octant between $z=0$ and $z=5$.

Soln:



Project the given surface in yz plane

$$\phi = x^2+y^2-16$$

$$\nabla\phi = 2x\hat{i} + 2y\hat{j}$$

$$|\nabla\phi| = \sqrt{4x^2+4y^2} = 2\sqrt{x^2+y^2} \qquad (\because x^2+y^2=16)$$

$$= 2\sqrt{16}$$

$$= 8$$

$$\therefore \hat{n} = \frac{\nabla\phi}{|\nabla\phi|} = \frac{2(x\hat{i}+y\hat{j})}{8}$$

$$= \frac{x\hat{i}+y\hat{j}}{4}$$

$$\vec{F}\cdot\hat{n} = (z\hat{i} + x\hat{j} + 3y^2z\hat{k}) \cdot \frac{x\hat{i}+y\hat{j}}{4}$$

$$= \frac{xz+xy}{4} = \frac{x(y+z)}{4}$$

$$\hat{n} \cdot \hat{i} = \frac{x\hat{i} + y\hat{j}}{4} \cdot \hat{i}$$

$$= \frac{x}{4}$$

$$\therefore \iint_S \vec{F} \cdot \hat{n}\, ds = \iint_R \vec{F} \cdot \hat{n} \frac{dy\,dz}{|\hat{n} \cdot \hat{i}|}$$

$$= \int_{z=0}^{z=5} \int_{y=0}^{y=4} \frac{x}{4}(y+z) \frac{dy\,dz}{\frac{x}{4}}$$

$$= \int_{z=0}^{z=5} \int_{y=0}^{y=4} (y+z)\, dy\, dz$$

$$= \int_{z=0}^{z=5} \left[\frac{y^2}{2} + yz\right]_0^4 dz$$

$$= \int_0^5 (8+4z)\, dz$$

$$= \left[8z + 2z^2\right]_0^5$$

$$= 40 + 50$$

$$= 90 \text{ //}$$

③ If the velocity $\vec{v} = 2y\hat{i} + 2\hat{j} + xz\hat{k}$ m/sec, find the flux through the surface $y = x^2$, $0 \le x \le 3$, $0 \le z \le 2$.

Soln:- Let $\phi = x^2 - y$

$$\nabla\phi = 2x\hat{i} - \hat{j}$$

$$|\nabla\phi| = \sqrt{4x^2+1}$$

$$\hat{n} = \frac{\nabla\phi}{|\nabla\phi|} = \frac{2x\hat{i} - \hat{j}}{\sqrt{4x^2+1}}$$

Taking the projection in xz plane we get

$$ds = \frac{dx\,dz}{|\hat{n}\cdot\hat{j}|}$$

$$\therefore |\hat{n}\cdot\hat{j}| = \frac{2x\hat{i} - \hat{j}}{\sqrt{4x^2+1}} \cdot \hat{j}$$

$$= \left| \frac{-1}{\sqrt{4x^2+1}} \right|$$

$$= \frac{1}{\sqrt{4x^2+1}}$$

$$\therefore \iint_S \vec{F}\cdot\hat{n}\,dS = \iint_R (y\hat{i} + 2\hat{j} + xz\hat{k}) \cdot \frac{2x\hat{i} - \hat{j}}{\sqrt{4x^2+1}} \cdot \frac{dx\,dz}{\frac{1}{\sqrt{4x^2+1}}}$$

$$= \int_{x=0}^{x=3} \int_{z=0}^{z=2} (2xy - 2)\,dx\,dz$$

$$= 2\int_{x=0}^{x=3} \int_{z=0}^{z=2} [x(x^2) - 1]\,dz\,dx$$

$$= 2\int_0^3 [x^3 z - z]_0^2\,dx$$

$$= 2\int_0^3 (2x^3 - 2)\,dx$$

$$= 4\int_0^3 (x^3 - 1)\,dx$$

$$= 4\left[\frac{x^4}{4} - x\right]_0^3$$

$$= 4\left[\frac{81}{4} - 3\right]$$

$$= 81 - 12$$

$$= 69 \text{ m}^3/\text{sec}.$$

(4) Evaluate $\iint_S (yz\hat{i} + xz\hat{j} + xy\hat{k}) \cdot \vec{ds}$ where S is the surface of the sphere $x^2+y^2+z^2=a^2$ in the first octant.

Soln:- Let $\phi = x^2+y^2+z^2-a^2$

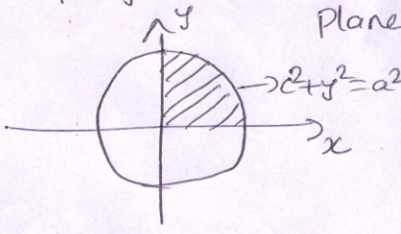
$$\nabla\phi = 2x\hat{i} + 2y\hat{j} + 2z\hat{k}$$

$$|\nabla\phi| = \sqrt{4x^2+4y^2+4z^2}$$

$$= 2\sqrt{a^2}$$

$$= 2a \qquad (\because x^2+y^2+z^2=a^2)$$

Projection in xy Plane



$$\vec{F} \cdot \hat{n} = (yz\hat{i} + xz\hat{j} + xy\hat{k}) \cdot \frac{x\hat{i}+y\hat{j}+z\hat{k}}{a}$$

$$= \frac{xyz + xyz + xyz}{a}$$

$$= \frac{3xyz}{a}$$

$$\therefore \iint_S \vec{F}\cdot\hat{n}\,ds = \iint_R \vec{F}\cdot\hat{n}\,\frac{dx\,dy}{|\hat{n}\cdot\hat{k}|}$$

$$= \int_0^a \int_0^{\sqrt{a^2-x^2}} \frac{3xyz}{a}\,\frac{dx\,dy}{\frac{z}{a}}$$

$$= 3\int_0^a \int_0^{\sqrt{a^2-x^2}} xy\,dx\,dy$$

$$= 3\int_0^a \left.\frac{xy^2}{2}\right|_0^{\sqrt{a^2-x^2}} dx$$

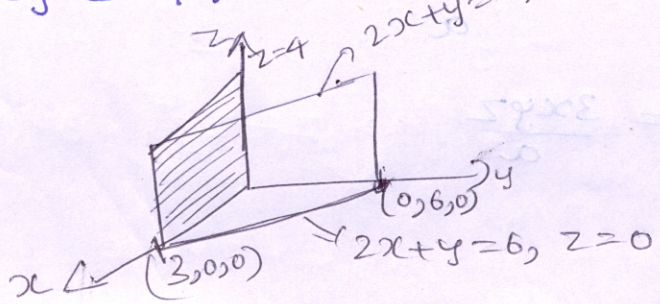
$$= \frac{3}{2}\int_0^a x(a^2-x^2)\,dx$$

$$= \frac{3}{2}\left[\frac{a^2x^2}{2} - \frac{x^4}{4}\right]_0^a$$

$$= \frac{3}{2}\left[\frac{a^4}{2} - \frac{a^4}{4}\right]$$

$$= \frac{3a^4}{8}$$

(5) Evaluate $\iint_S \vec{F}\cdot\hat{n}\,ds$ where $\vec{F} = y\hat{i} + 2x\hat{j} - z\hat{k}$ and S is the surface of the plane $2x+y=6$ in the first octant cut by $z=4$.



Let $\phi = 2x+y-6$

$$\nabla\phi = 2\hat{i}+\hat{j}$$

$$|\nabla\phi| = \sqrt{4+1} = \sqrt{5}$$

$$\therefore \hat{n} = \frac{2\hat{i}+\hat{j}}{\sqrt{5}}$$

$$\vec{F}\cdot\hat{n} = (y\hat{i}+2x\hat{j}-2\hat{k})\cdot\frac{2\hat{i}+\hat{j}}{\sqrt{5}}$$

$$= \frac{2y+2x}{\sqrt{5}}$$

Project on to xz plane, we have

$$ds = \frac{dx\,dz}{|\hat{n}\cdot\hat{j}|} = \frac{dx\,dz}{\frac{1}{\sqrt{5}}}$$

$$\iint_S \vec{F}\cdot\hat{n}\,ds = \iint_R \frac{2y+2x}{\sqrt{5}} \times \frac{dx\,dz}{\frac{1}{\sqrt{5}}}$$

$[\because 2x+y=6$
$\Rightarrow y = 6-2x$

$$= \int_{x=0}^{x=3}\int_{z=0}^{z=4} [2(6-2x)+2x]\,dz\,dx$$

$$= \int_{x=0}^{x=3}\int_{z=0}^{z=4} (12-2x)\,dz\,dx$$

$$= \int_{x=0}^{x=3} (12z-2xz)\Big|_0^4 dx$$

$$= \int_0^3 (48-8x)\,dx$$

$$= [48x-4x^2]_0^3 = 144-36$$

$$= 108 \text{ //}$$

(1) Evaluate $\iint_S \vec{F}\cdot\hat{n}\,ds$ where $\vec{F} = 6z\hat{i} - 4\hat{j} + y\hat{k}$ and S is the portion of the plane $2x+3y+6z=12$ which is in the first octant.

Answer:- 8

(2) Evaluate $\iint_S \vec{F}\cdot\hat{n}\,ds$ where $\vec{F} = (x+y^2)\hat{i} - 2x\hat{j} + 2yz\hat{k}$ and S is the surface of the plane $2x+y+2z=6$ in the first octant.

Answer:- 81

(3) Evaluate $\iint_S \vec{F}\cdot\hat{n}\,ds$ if $\vec{F} = z\hat{i} + x\hat{j} + y\hat{k}$ and S is the surface of the cylinder $x^2+y^2=1$ contained in the first octant between the planes $z=0$ and $z=2$.

Answer:- 3